

Octal to Binary & Binary to Octal Conversion

- >> There are eight different digits in octal number system.
- >> Each one of them can be represented by a equivalent 3-bit number.

$$r = 8$$

$$0 \text{ to } (8-1)$$

$$0 \text{ to } 7$$

0 -	0 0 0
1 -	0 0 1
2 -	0 1 0
3 -	0 1 1
4 -	1 0 0
5 -	<u>1 0 1</u>
6 -	1 1 0
7 -	1 1 1

Octal to Binary:

i) $(37.45)_8 \longrightarrow (011111.100101)_2$

Diagram showing the conversion of each octal digit to its 3-bit binary equivalent:

$3 \rightarrow 011$, $7 \rightarrow 111$, $4 \rightarrow 100$, $5 \rightarrow 101$

ii) $(22.07)_8 \longrightarrow (010010.000111)_2$



T - 111

Binary to Octal:

H.W. i) $(672.13)_8 \rightarrow (\quad)_2$ i) $(010110.110)_2 \rightarrow (26.6)_8$

$\underbrace{}_2 \quad \underbrace{}_1 \quad \uparrow \quad \underbrace{}_2$
 $\longleftarrow \longrightarrow$

ii) $(10111011.1001)_2 \rightarrow (\quad)_8$ ii) $(001001001.100)_2 \rightarrow (011.4)_8$

$\underbrace{}_1 \quad \underbrace{}_1 \quad \underbrace{}_1 \quad \underbrace{}_2$

1. 110111010.001011

2. 273.44